

2018 JC2 H2 BIO P2 PRELIM EXAM ANSWER SCHEME

Answer **all** questions.

- 1 Fig. 1.1 shows an electron micrograph of a pancreatic cell that secretes large amounts of insulin that helps to regulate blood glucose level.

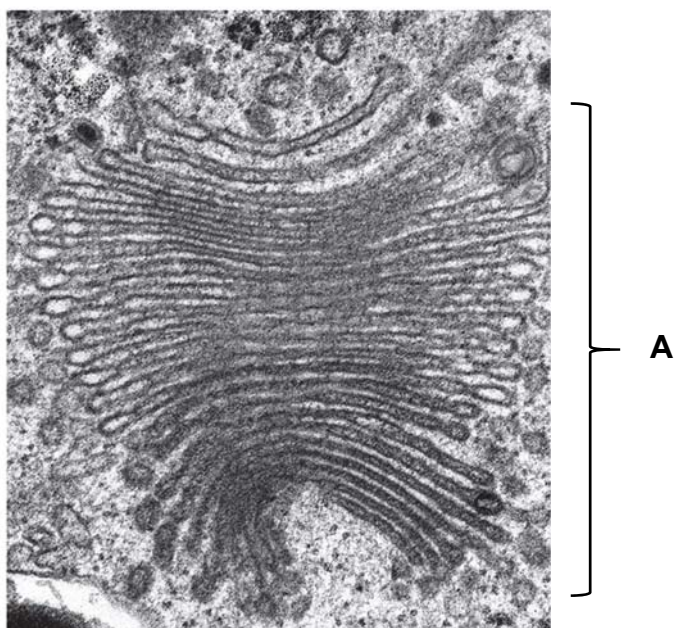


Fig. 1.1

(a) With reference to Fig. 1.1,

- (i) identify organelle **A**;

Golgi apparatus/ body;

[1]

- (ii) describe **two** identifying features of organelle **A** that allows its identification in (a)(i).

1. a stack of discrete, flattened, memb-bound sacs called cisternae;

2. distinct memb with 2 faces cis/ convex & trans/ concave);

3. vesicles pinching & joining from ends of each sac;

(any 2)

[2]

Fig. 1.2 shows a diagram of the molecular structures of tristearin (a triglyceride) and phosphatidylcholine (a phospholipid).

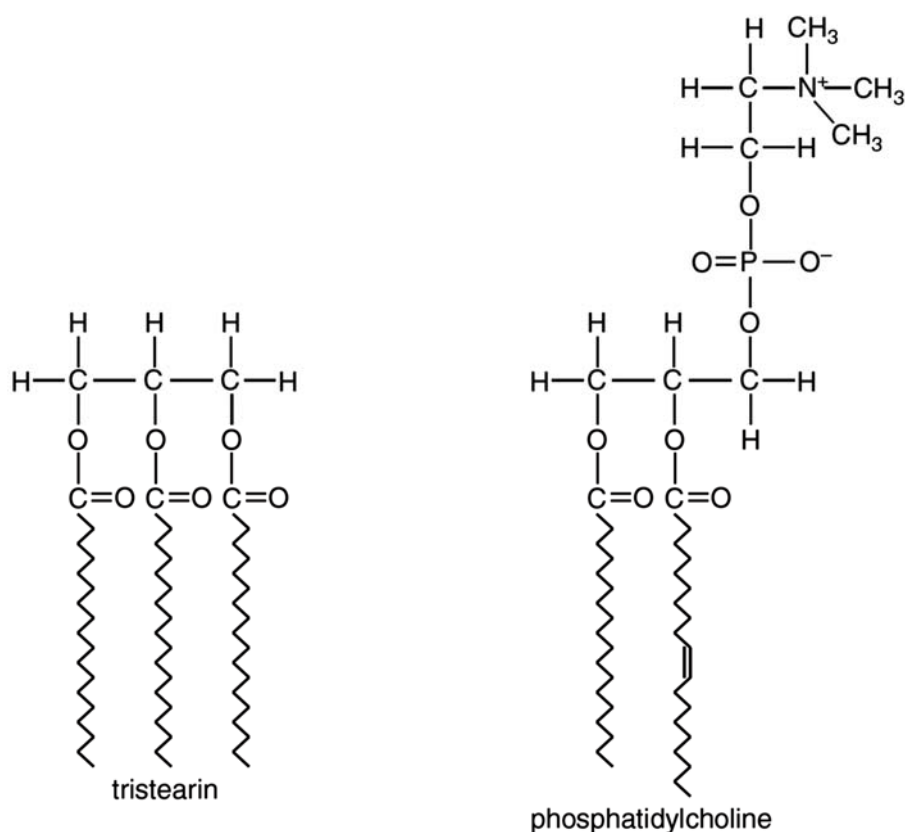


Fig. 1.2

(b) Table 1 shows a structural difference between the two molecules shown in Fig. 1.2.

Complete Table 1 with two further **structural** differences **other than** in numbers of different types of atoms.

Table 1

structural feature	tristearin	phosphatidylcholine
length of fatty acid chain	all same length	different lengths
degree of saturation	all FA tails are saturated	1 FA tail is unsaturated / have C=C
functional grp	glycerol & carboxylic acid	glycerol, carboxylic acid & phosphate
presence of phosphate group	absent	present
no. of FA	3	3

- (c) Triglyceride is used as energy storage while phospholipids are membrane components. Explain why phospholipids are suitable membrane components but not triglyceride.

1. amphipathic

has hydrophilic head and hydrophobic tail;

2. able to form bilayer

w hydrophilic heads interxt w aq medium in/s & o/s cell (while forming a hydrophobic core that prevents free movement of hydrophilic subst);

[2]

Cells in the pancreas secrete enzymes, such as amylase and trypsin, into a duct. The enzymes are packaged in vesicles so that they can be exported from these cells as shown in Fig. 1.3.

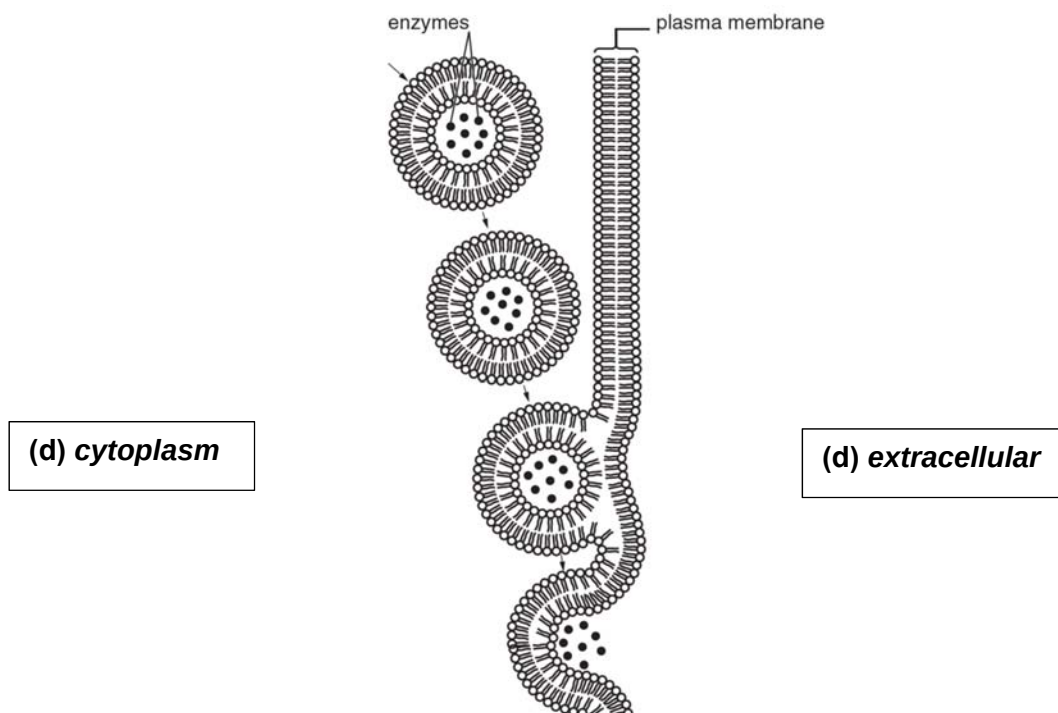


Fig. 1.3

- (d) On Fig. 1.3, label the cytoplasm of the cell as '**cytoplasm**' and extracellular fluid as '**extracellular**'. [1]

(e) With reference to Fig. 1.3,

(i) explain how enzymes that are secreted by cells in the pancreas are packaged into vesicles and exported, after their synthesis at the endoplasmic reticulum.

1. enz packaged in tpt vesicles pinch off from surface of rER

pinch off from surface of rER;

2. tpt vesicles travel thru cytoplasm / or idea of transport

fuse with A/ Golgi apparatus at cis face;

3. protein modification occurs (adding phosphate grp/ adding, deleting, substituting sugar monomers)

modification results in molecular identification tag on enz;

4. vesicles containing enz pinch off

from trans face of A/ Golgi apparatus;

5. vesicles translocate towards CSM

via cytoskeleton (ATP req'd);

6. memb vesicle fuses with CSM

enz released out of cell via exocytosis;

[6]

(ii) explain **one** property of the plasma membrane that allows vesicle formation.

1. memb is fluid

allows evagination of memb to form vesicles;

2. phospholipid molecules are held by weak hydrophobic interxns

thus capable of lateral movement (within the monolayer);

[2]

(f) Describe **two** advantages of having plasma membranes **within** the cell.

1. allows compartmentalisation

to setup unique / optimum conditions for biochem rxns (e.g. acidic pH in lysosome);

2. regulates movement of subst in and out of cell / organelle

(by having prot trspters that only allows movement of specific subst);

[2]

[Total: 18]

2 Fig. 2.1 shows a bacterium cell.

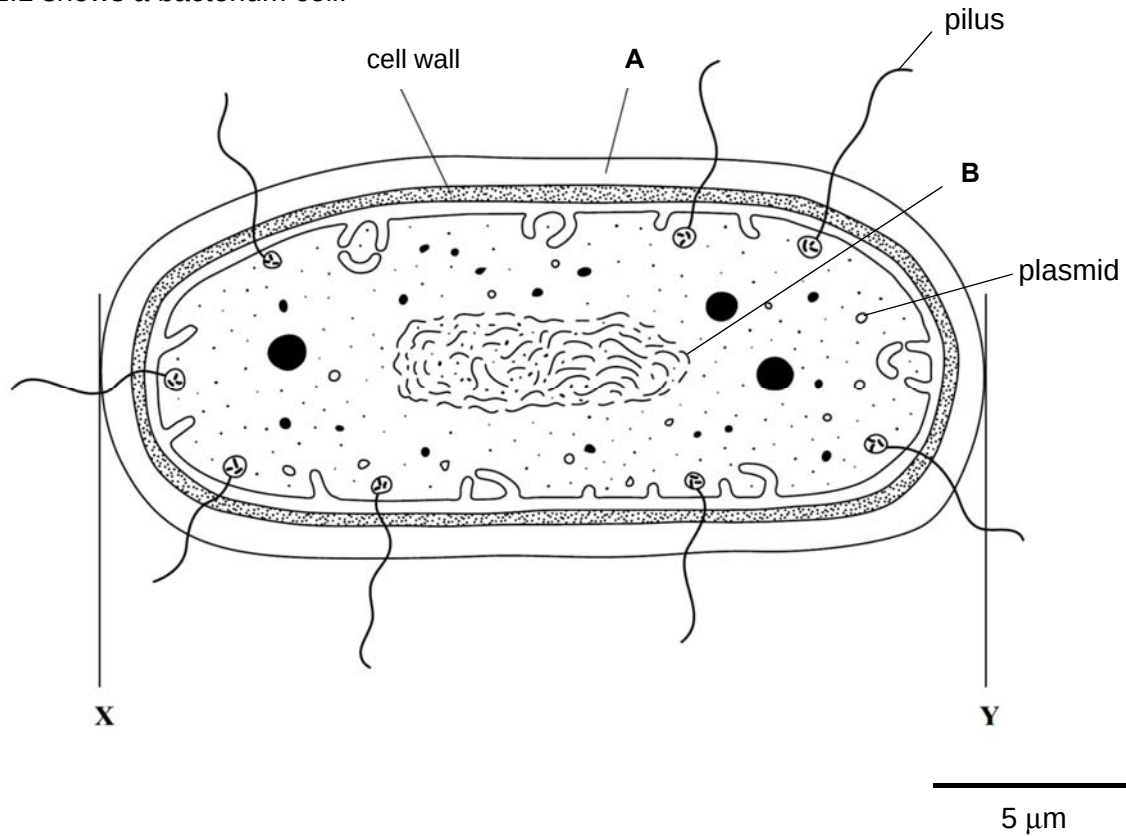


Fig. 2.1

(a) Identify structures **A** and **B**.

A *capsule;*

B *DNA / nucleoid / chromosome;*

[2]

(b) Calculate the length of the bacterium cell from **X** to **Y** in μm . Show your working clearly.

- *Accept measurement of line between 2.5 – 2.7 cm*
- *Accept measurement of XY between 11.5 – 11.7 cm*
- *Accept answers with rounding off calculations*

[1]

Bacterial cells can acquire new alleles through genetic transfers between bacterial cells. One way is through plasmids called F plasmids in bacterial cells, another is through bacteriophages.

(c) Describe how a bacterium cell can acquire new alleles through F plasmids.

1. F plasmid in F+ cell syn sex pilus that binds to F- cell

retracts to form cytoplasmic bridge;

2. enz cleaves F plasmid at oriT

s/s transferred from F+ to F- cell;

3. F plasmid replicated by (rolling circle mechanism)

s/s F plasmid acts as template to syn 2nd F plasmid strand, F- cell becomes F+ with new alleles found on F plasmid;

[3]

An investigation was conducted on a temperate phage that infects a new strain of bacteria, to find the prophage insertion site on the bacterial chromosome.

Fig. 2.2 shows the chromosome map of the bacterial strain. The chromosome map shows positions of the genes *ade*, *arg*, *cys*, *his*, *leu* and *pro*, which are each involved the synthesis of amino acids essential for bacterial growth.

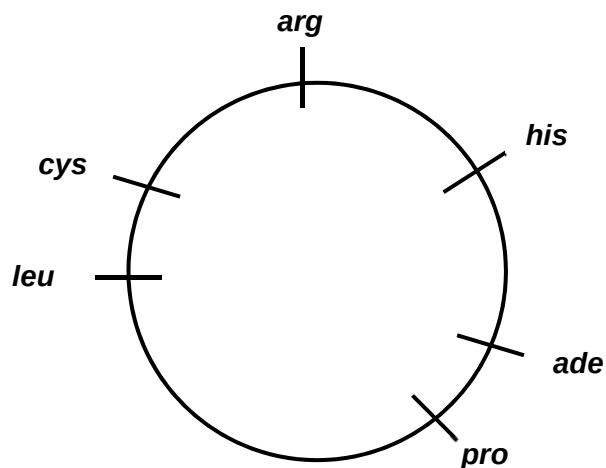


Fig. 2.2

The bacterial strain of genotype *ade*⁺ *arg*⁺ *cys*⁺ *his*⁺ *leu*⁺ *pro*⁺ (where '+' indicates functional wild-type allele) was used as a source of the phage. The phages were then added to a bacterial strain of genotype *ade*⁻ *arg*⁻ *cys*⁻ *his*⁻ *leu*⁻ *pro*⁻ (where '-' indicates non-functional allele).

After a short incubation, samples of these bacteria were plated on six different media. The plates were then observed for growth of bacterial colonies. Table 2 shows the results of the experiment.

Table 2

medium	nutrient supplementation in medium						presence of colonies
	ade	arg	cys	his	leu	pro	
1	✕	✓	✓	✓	✓	✓	N
2	✓	✕	✓	✓	✓	✓	N
3	✓	✓	✕	✓	✓	✓	C
4	✓	✓	✓	✕	✓	✓	N
5	✓	✓	✓	✓	✕	✓	C
6	✓	✓	✓	✓	✓	✕	N

Key:

‘✓’ indicates presence of amino acid

‘✕’ indicates absence of nutrient supplement

‘C’ indicates presence of colonies,

‘N’ indicates absence of colonies

(d) With reference to Table 2,

(i) indicate on Fig. 2.2, the position of prophage insertion. [1]

1. accept indication of insertion point between cys and leu;

(ii) explain why colonies are able to grow on medium 3 and 5.

2. temperate phage infects donor bact cell

integrated prophage b/w cys+ and leu+ into bact chr;

3. cys+ and leu+ next to prophage accidentally excised & packaged

when temperate phage switch from lysogenic to lytic cycle;

4. phage infects recipient bact cell deficient in all enz

homologous recomb of cys & leu genes occurs resulting in recipient bact now with cys+ & leu+ alleles;

[3]

[Total: 10]

- 3 Fig. 3.1 shows the structure of a mature tRNA that is involved in the production of a globular enzyme in a eukaryotic cell.

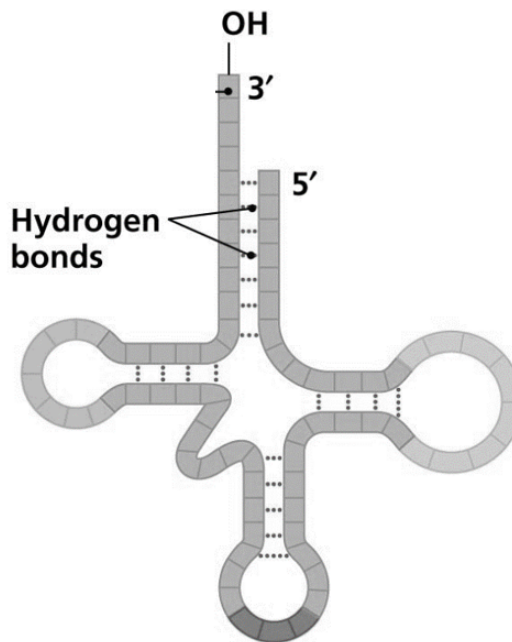


Fig. 3.1

- (a) With reference to Fig. 3.1, explain the roles of hydrogen bonds in the proper functioning of tRNA.

1. *H-bds are formed b/w compl bp allows folding of tRNA into (specific) 3D conformation / looped struc / clover-shape*

for stability of tRNA (during activation & translation);

2. *idea of folding, to fit into compl binding site of large ribosome subunit (P & A site);*

3. *idea of folding, to fit into active site of aminoacyl-tRNA synthase for tRNA activation;*

4. *H-bds formed b/w anticodons on tRNA with compl codons on mRNA;*

[3]

- (b) The length of the tRNA gene is longer than that of the mature tRNA.

Outline how a **mature** tRNA molecule is synthesised in eukaryotes.

1. *[initiation] RNA poly binds to promoter of tRNA gene to transcribe tRNA*

together with basal TFs to form transcription initiation complex;

2. *[elongation] adding of compl ribonucleotides from template to form pre-tRNA*

modification of bases (addition of CCA stem) on pre-tRNA;

3. *introns are excised & exons are spliced together*

by spliceosomes to form mature tRNA;

[3]

Fig. 3.2 shows a pathway in the regulation of gene expression to synthesise enzymes in a eukaryotic cell can be achieved.

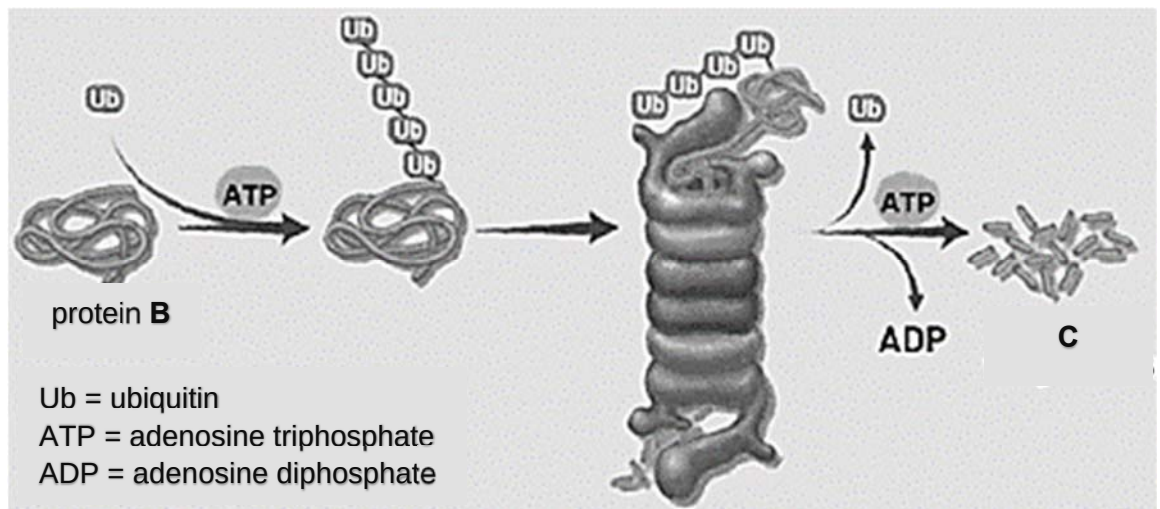


Fig. 3.2

(c) With reference to Fig. 3.2,

(i) name the process involved;

ubiquitination/ post translational modification/ proteolytic degradation;

[1]

(ii) explain how product **C** is formed from protein **B**.

1. prot selected for degradation are tagged with/ bind to multiple ubiquitin mols

ATP is req'd in tagging process;

2. target prot tagged with ubiquitin enter proteasome

enz of proteasome hydrolyse peptide bonds;

3. into small peptide fragments

ubiquitin mols are released & re-used;

[3]

[Total: 10]

- 4 Crossing over between homologous chromosomes is an important process that gave rise to genetic variation.

Errors during crossing over can result in abnormal chromosomes that will affect the phenotype of an organism.

Fig. 4.1 show the result of an unequal crossing over event in prophase I.

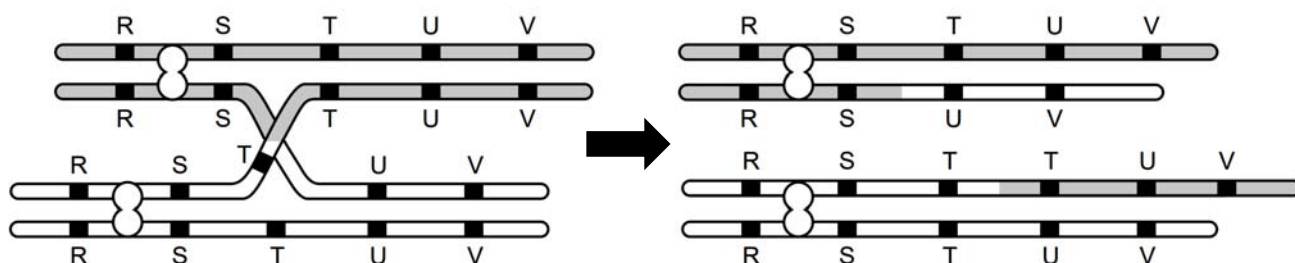


Fig. 4.1

- (a) (i) State the type of mutation that has occurred in Fig. 4.1.
- duplication of chr segment T / deletion of chr segment T;*** [1]
- (ii) Suggest the possible effects on the phenotype of the organism if protein T is a hormone involved in metabolism.
- duplication***
- 1. more prot T pdc due to more mRNA transcribed from DNA template;***
- 2. higher metabolism > normal; OR***
- deletion***
- 3. no prot T pdc due to no gene T transcribed;***
- 4. lower metabolism < normal;*** [2]
- gene T is a dominant allele, one copy is sufficient for expression into phenotype***
(Any pair of answers)

Fig. 4.2 shows the human karyotype of one of the most common chromosomal abnormalities. The affected embryos or foetuses with this condition ended up in miscarriages in the first trimester.

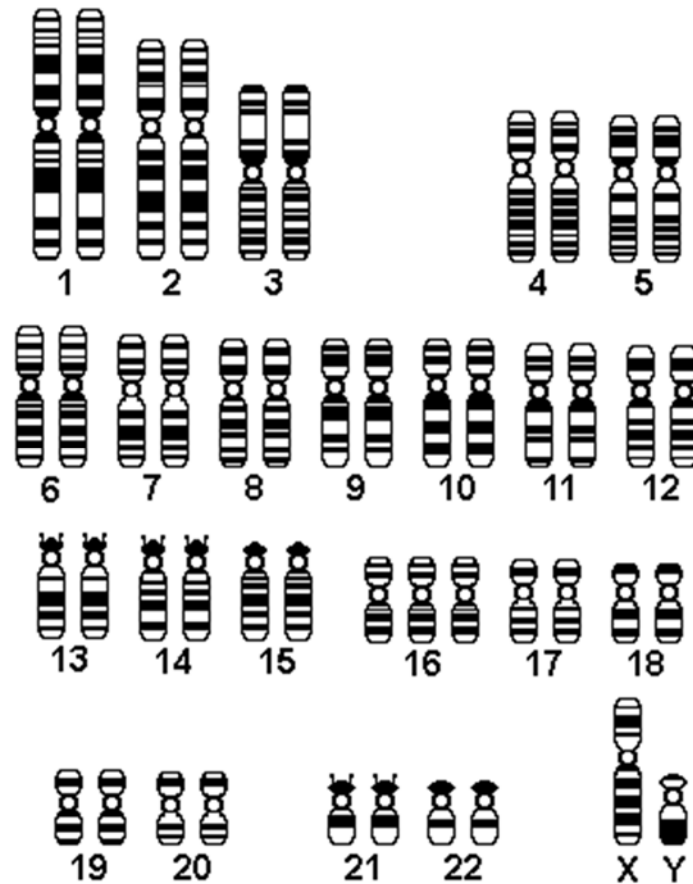


Fig. 4.2

(b) With reference to Fig. 4.2, explain how this condition arises.

1. **trisomy 16 / w.t.e.;**

2. **aneuploidy**

due to unequal separation of homolog chr (during anaphase I);

3. **errors in spindle formation**

resulted in one (haploid) gamete with 2 copies of chr 16;

4. **fusion with a normal haploid gamete**

restores diploid condition for all other chr pairs (except for chr 16);

[4]

- (c) Fig. 4.1 and Fig 4.2 are examples of chromosome aberration.

Use examples other than the ones shown, outline what does *chromosome aberration* mean.

1. changes to struc & nos of chr;

2. struc aberration can be due to translocation

exchange of chr segment b/w non-homolog chr; OR

3. struc aberration can be due to inversion

part of chr is cut out, reversed & reinserted back;

4. numerical aberration (of chr sets)

unequal separation of chr sets resulting in polyploidy;

[3]

[Total: 10]

- 5 Two independently assorting genes coding for eye colour and the presence of wing crossveins in the *Drosophila* fruit fly were investigated in a few genetic crosses with different adults.

The outcomes of four crosses are shown below in Table 5.

Table 5

Cross	Parental phenotypes	Orange, crossveins present	Orange, crossveins absent	Red, crossveins present	Red, crossveins absent
1	orange, crossveins present, orange, crossveins present	83	26	0	0
2	red, crossveins present, red, crossveins absent	20	18	65	63
3	red, crossveins absent, red, crossveins present	0	0	71	81
4	red, crossveins present, red, crossveins present	28	11	93	34

(a) Using suitable symbols,

- (i) deduce the genotypes of the adults in Cross 2.

Red, crossveins present

RrPp;

Red, crossveins absent

Rrpp;

[2]

- (ii) explain why Cross 1 and Cross 3 have zero offspring in some phenotypic categories.

Cross 1

1. orange, recessive trait coded by *rr*

pure-breeding parents for eye colour;

Cross 3

2. red, dominant trait expressed with 1 allele copy

at least one parent has both copies of dominant alleles (RR) for eye colour;

[2]

- (b) In order to test the significance of differences between observed and expected results, a statistical test was carried out for Cross 4.

- (i) Name the statistical test carried out.

Chi squared test;;

[1]

- (ii) State the expected offspring ratio.

9: 3: 3: 1;;

[1]

- (iii) Explain what does it mean when the results of the chi squared test has a probability of more than 0.05.

1. no sig diff b/w observed & expected results

genes assort independently & segregate randomly into gametes;

2. genes for eye colour & gene for presence of crossveins are not linked OR due to occurrence of chanced event (e.g. random fertilization, small sample size etc); [2]

- (c) Using a named example, describe how the environment may affect the phenotype of organisms.

1. temp affecting coat colour of Himalayan rabbits;

black coat at temp <33°C due to melanin pdtn by enzyme; (A) converse argument

2. prot content in diet affects devt of larvae in honey bees;

larvae fed with royal jelly continuously dev mature reproductive system & dev into the queen; (A) converse argument

3. AVP (state e.g.);

AVP (brief descriptions); [2]

[Total: 10]

6 Fig. 6.1 shows a section of a chloroplast.

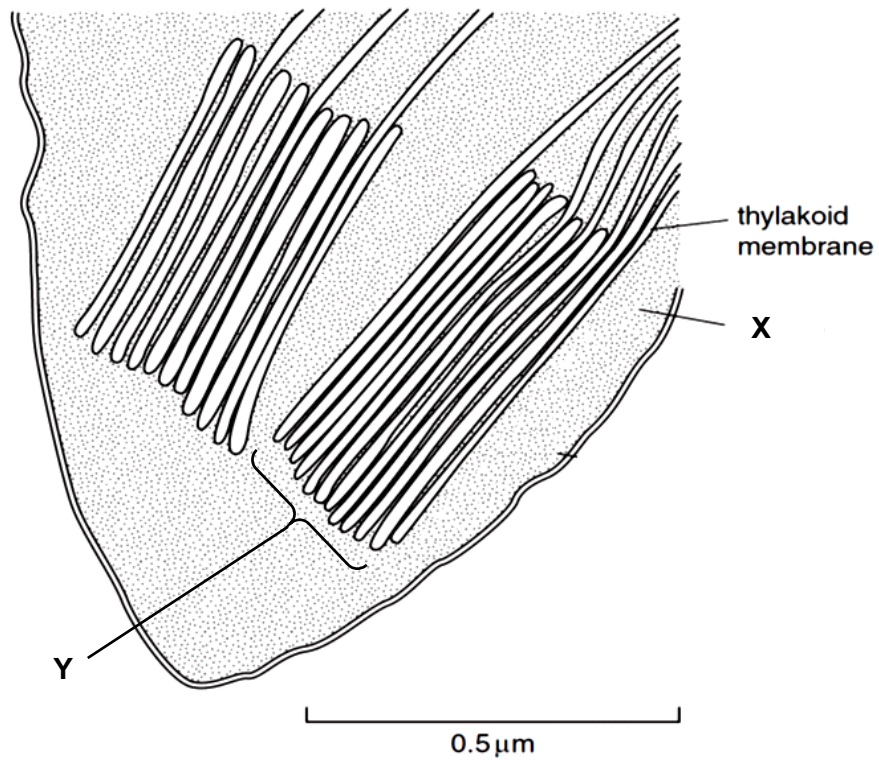


Fig. 6.1

(a) Identify structures X and Y.

X **stroma;**

Y **granum;**

[2]

Fig. 6.2 shows ATP synthase embedded in the thylakoid membrane.

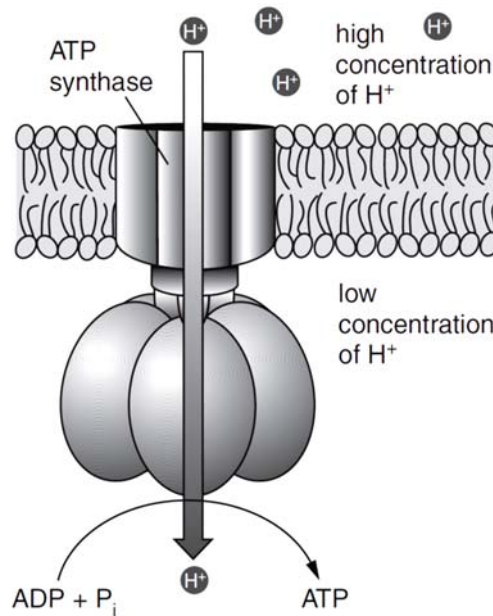


Fig. 6.2

(b) (i) Describe how the presence of light led to the synthesis of ATP.

1. *photon of light accepted by PSII / chl a*

cause photoactivation of chl a in rxn ctr / photoexcitation of e- in chl a;

2. *e- is passed down on a series of e- carriers in ETC (to PSI)*

of progressively lower energy level;

3. *energy lost (during e- transfer) is used to pump/against conc grad H⁺ ions from stroma to thylakoid lumen
accumulation of H⁺ in thylakoid lumen creates a proton grad / motive force;*

4. *H⁺ ions diffuse from thylakoid lumen back into stroma via (Fo of) ATP synthase
(F₁) catalyses phosphorylation of ADP → ATP, ATP synthesised via chemiosmosis;*

[4]

(ii) Describe the fate of ATP produced in Fig. 6.2.

1. *provide en for Calvin cycle*

reduction phase, GP → TP;

2. *and phosphate grp*

RuBP regeneration, TP → RuBP;

[2]

Fig. 6.3 shows the relationship between CO₂ assimilation rate and increasing light intensity in a plant.

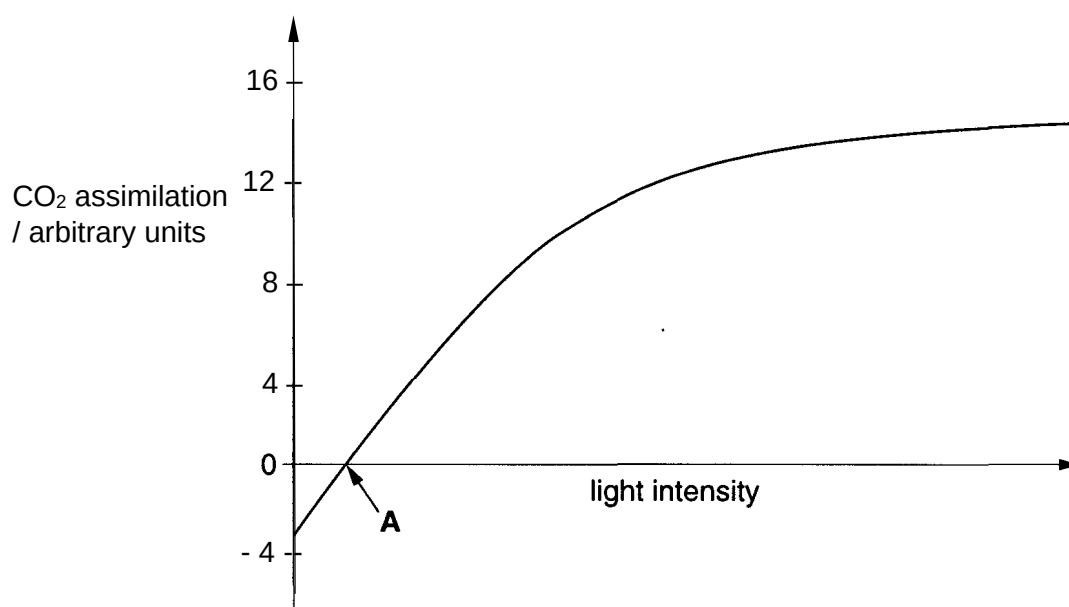


Fig. 6.3

(c) With reference to Fig. 6.3,

- (i) State the total volume of CO₂ used by the plant for photosynthesis when light intensity is no longer the limiting factor.

14 + 3 = 17 arbitrary units;

[1]

- (ii) Describe the significance of point **A** to the growth of the plant.

1. A is the (light) compensation point

where PS rate = resp rate;

2. no net gain in dry mass thus no growth

as products of PS (glucose & O₂) are used up for resp and products of resp (H₂O & CO₂) are used up for PS;

[2]

[Total: 11]

- 7 Fig. 7.1 shows the structure of the hormone, insulin, which is responsible for the regulation of blood glucose level back to its normal set point.

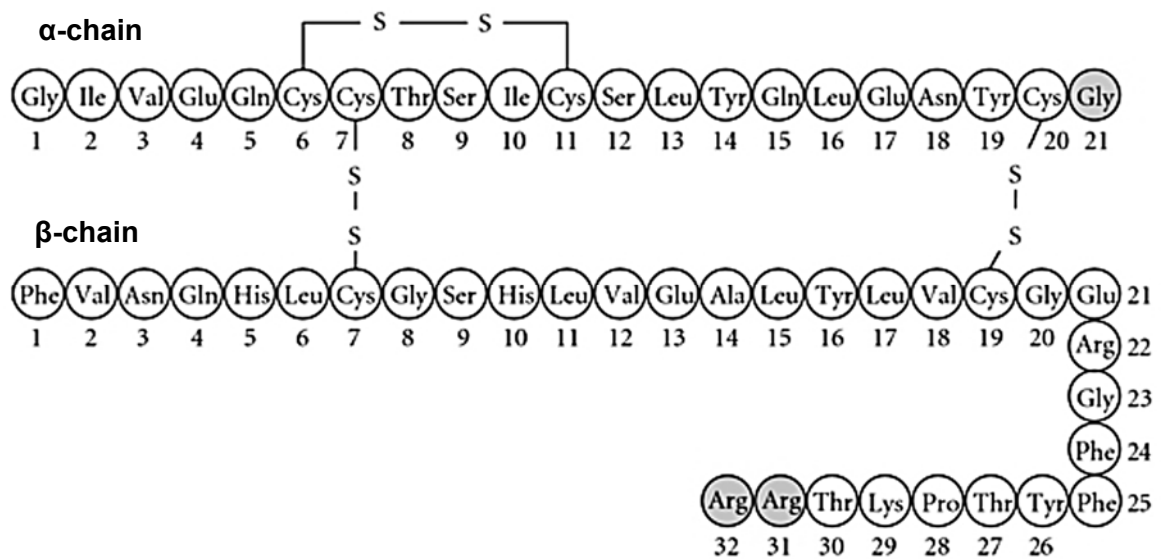


Fig. 7.1

- (a) With reference to Fig. 7.1, describe the structure of insulin.

1. made up of 2 polypeptide chains

α chain with 21 aas and β chain with 32 aas / 53 aas in its pri prot struct

OR aas linked by peptide bonds;

2. presence of disulfide bridges

b/w (sulfhydryl grps) of cys residues;

3. (intrachain) within α -chain b/w cys at position 6 & 11;

4. (interchain) b/w chains at cys at position 7 in α -chain & position 7 in β -chain

OR cys at position 20 in α -chain & position 19 in β -chain;

[3]

Metformin is used as a drug to treat hyperglycaemic conditions in patients with Type II diabetes where the skeletal muscles are resistant to insulin stimulation. It is transported into the cell by specific protein carrier.

Fig. 7.2 illustrates how metformin influences the cellular activity involved in the signal transduction pathway. AMPK is a kinase that is involved in energy sensing and is activated by AMP (adenosine monophosphate) which is one of the products of ATP hydrolysis.

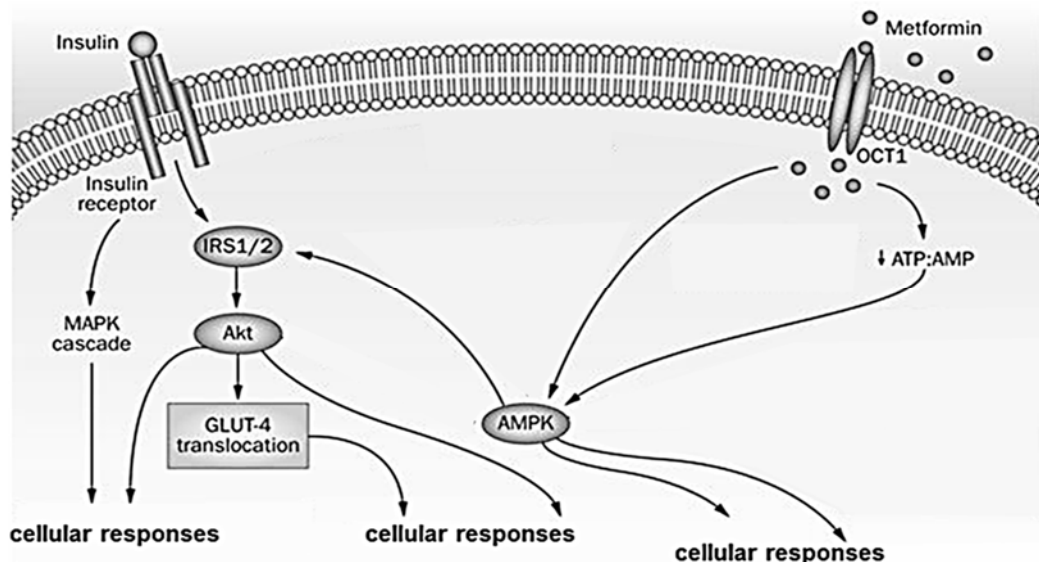


Fig. 7.2

(b) With reference to Fig. 7.2,

(i) explain how metformin can be used to decrease the blood glucose level in patients with Type II diabetes.

1. metformin enters cell via (prot channel / trspter) OCT1

by facilitated diffusion / which provides hydrophilic channel;

2. leads to ↓ ATP:AMP (by increasing ATP hydrolysis) / ↑ AMP

which activates AMPK;

3. AMPK phosphorylates & activates IRS1/2

which activates Akt;

4. activated Akt leads to ↑ GLUT-4 translocation to the CSM

leading to ↑ nos. of GLUT-4 on CSM ➔ ↑ glucose uptake into cell;

5. idea of activation of other cellular responses by Akt, AMPK

e.g. ⊕ glycogen synthase to ↑ glycogen synthesis

e.g. ⊖ glycogen phosphorylase to ↓ glycogen hydrolysis;

[4]

- (ii) Akt is known to stimulate other cellular responses in the insulin signalling pathway.

Suggest how activation of Akt can lead to different cellular responses.

1. acts as relay prot / 2nd messenger;

2. that can bind to / activate diff relay prot / kinases (to cause phosphorylation cascade in other signalling pathways) ;

3. leading to activation / inactivation of diff enz that can catalyse diff cellular rxns / cite e.g.;

4. acts as TFs that can up / downregulate expression of diff genes;

[3]

[Total: 10]

- 8 A mammoth can be any species belonging to the extinct genus *Mammuthus*, proboscideans commonly equipped with long, curved tusks. They were members of the family Elephantidae which contains, along with mammoths, the two genera of modern elephants and their ancestors.

About three million years ago, the ancestors of mammoths migrated from Africa into Europe and Asia. There, about 1.7 million years ago, the steppe mammoth evolved and became adapted to the cooler conditions. Then, about 700 000 years ago, as the climate changed and the Arctic became much colder, the woolly mammoth evolved.

Woolly mammoths showed a number of obvious adaptations to reduce heat loss, including thick fur, small ears and small tails.

- (a) Explain how variation and natural selection may have brought about the evolution of the woolly mammoth from the steppe mammoth.
1. *steppe mammoth w phenotypic variation (due to presence of diff alleles / genetic variation / mutation)*
climate Δ (700 000 yr ago) colder temp act as selection pressure;
 2. *mammoth w favourable phenotype / adaptations to reduce heat loss selected for / have selective advantage*
e.g. thick fur, small ears and small tails;
 3. *such mammoths experience $\uparrow$ er survival rate & repro success*
pass down alleles coding for favourable phenotype to offspring lead to Δ in allele frequency in gene pool;
 4. *accumulate mutations in gene pool over time*
becomes genetically diverged from steppe mammoth ancestor $\rightarrow$ evolve to become woolly mammoth;

[4]

- (b) The phylogenetic relationship of the woolly mammoth, Asian elephant (*Elephas maximus*) and African elephant (*Loxodonta africana*) was then investigated.

Fig. 8.1 shows the phylogenetic tree obtained.

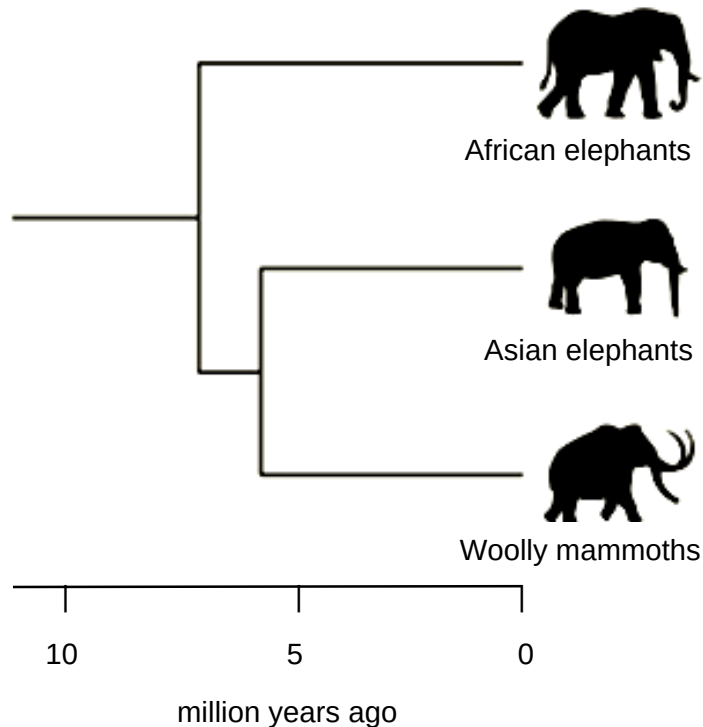


Fig. 8.1

With reference to Fig. 8.1, state

- (ii) when the African elephant diverged;

7.5 million years ago;

[1]

- (iii) when the Asian elephant diverged.

6 million years ago;

[1]

- (c) Explain how molecular methods can be used to construct the phylogenetic tree in Fig. 10.1.

1. select a common gene / prot among the 3 species

to compare the DNA / aa seq;

2. ↓ er no. of differences → share more recent common ancestor ORA

as they have less time to accumulate diff mutations / used to calculate length of time since divergence;

[2]

[Total: 8]

- 9 Cholera is an infectious disease caused by the bacterium *Vibrio cholerae*. The bacteria are transmitted between humans through the fecal-oral route. Contaminated food or water can cause infection which symptoms can occur very quickly, from a few hours to a few days.

Table 9 shows

- the economic status of each of five countries
- the number of cases of cholera reported to the World Health Organization (WHO) over a five year period for each country
- the population in 2006 and in 2010 of each country.

Table 9

country	economic status	number of cholera cases reported			population / millions	
		2006	2008	2010	2006	2010
Zimbabwe	low	789	60055	951	12.5297	12.5715
Uganda	low	5194	3726	2341	29.3703	33.4247
Angola	middle	67257	10511	1484	17.0104	19.0819
Cameroon	middle	922	0	10759	17.9484	19.5989
Canada	high	2	1	2	32.6490	34.1088

(a) With reference to Table 9,

- (i) suggest **two** reasons to explain the higher number of cholera cases reported in low economic status countries compared to high economic status country.

1. poor / inadequate sanitation / treatment of sewage;

2. no / poor water treatment → contaminated water;

3. poor access to healthcare

antibiotics / drugs / medication / vaccines not available;

4. less education about disease prevention / transmission or poor hygiene practices e.g. not washing hands;

[2]

(Any 2 points)

- (ii) suggest if economic status of a country is likely to be the main reason for the high number of cases of cholera.

1. *no*

quote any country with middle economic status e.g. Angola (67257 in 2006) having higher no. of cases than country with low status e.g. Uganda (5194 in 2006);

2. *other reasons like war / famine etc.;*

[2]

- (b) Cholera can be treated with antibiotics like tetracycline.

- (i) Suggest how tetracycline results in the death of the bacterium.

1. *inhibits prot synthesis OR
cell wall synthesis in bact*

*lack of essential metabolic enz causing bact cell death; OR
bact cell undergoes (osmotic) lysis due to weakened cell wall;*

2. *by binding to ribo subunit OR
transpeptidase*

*preventing formation of translation initiation cplx / translation OR
prevents cross linking of peptidoglycan in cell wall;*

[2]

- (ii) Explain why tetracycline does not cause death of human cells.

1. *humans have 80S ribo instead of 70S in bact with diff 3D config that
tetracycline cannot bind; OR*

human cells do not synthesize cell wall;

[1]

[Total: 7]

- 10 Dengue fever is a mosquito-borne viral disease and a regular epidemic in Thailand. The peak of the dengue epidemic period is around June to August during the rainy season. It is believed that climate is an important factor for dengue transmission.

Fig. 10.1 shows how changes in temperature affects the average monthly incidence rate of dengue cases in Chiang Rai, Thailand from 2004 – 2014.

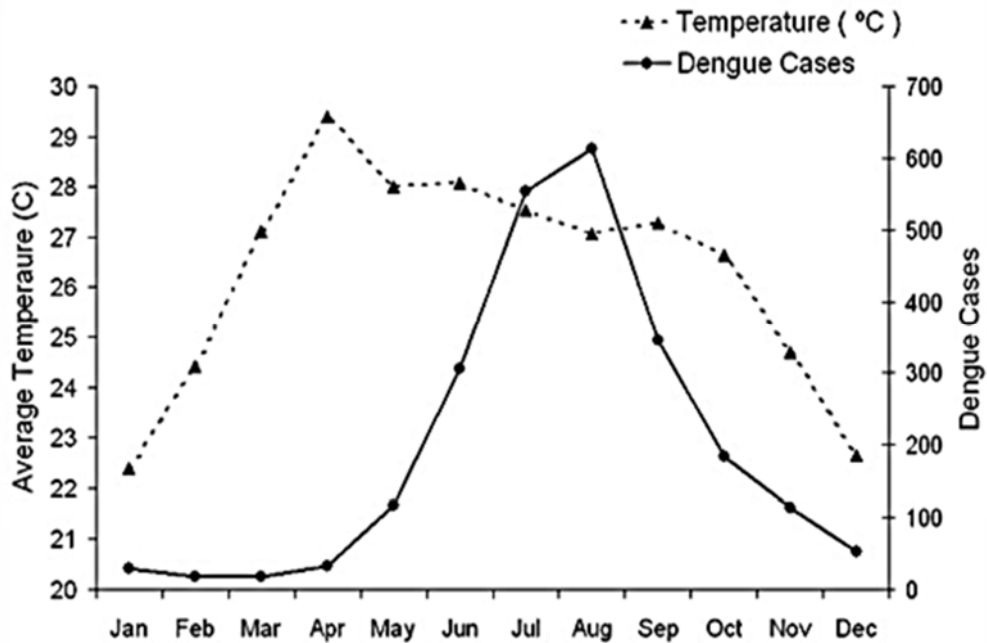


Fig. 10.1

- (a) State how each data point in Fig. 10.1 is obtained.

average calculation of all temp in same month from 2004 – 2014;

[1]

- (b) With reference to Fig. 10.1,

- (i) describe the trend of incidence of dengue cases with regard to changes in temperature;

1. temp ↑ sharply from 22 – 29.5 °C from Jan to Apr

almost constant no. of dengue cases at 20 cases;

2. temp ↓ from 29.5 – 23 °C from Apr to Dec

steep ↑ in dengue cases from 20 – 620 from Apr to Aug

steep ↓ in dengue cases from Aug to Dec from 620 – 50;

[2]

(ii) explain the trend described in (b)(i).

1. $\uparrow$ in ave temp $\uparrow$ metabolic activity in mosquitoes

$\uparrow$ in freq of effective collisions b/w E & S $\Rightarrow$ more ES cplx formed;

2. faster growth rate $\Rightarrow$ faster maturation rate

shorter time needed to complete life cycle;

3. more adult mosquitoes available to become vectors for dengue virus to infect more human host leading to $\uparrow$ no. dengue cases

observed in subsequent months after $\uparrow$ in temp due to time needed for mosquito to grow and mature; [3]

[Total: 6]